El Bambú: Phase 1

The first in a series of interventions on the Los Mollejones' indoor soccer court to amplify existing community spaces

Sustainable Buildings, Colorado State University & University of Costa Rica

Project Summary

Figure 1. Mollejones' community logo.

Mollejones is a beautiful rural community in Turrialba, Costa Rica, with "La Cancha" serving as the heart of the community. This central gathering space is a hub for all residents, fostering connections and activities. Using our core principles of sustainability, feasibility, collaboration, education, and longevity, we identified an opportunity to enhance this beloved area. Living with host families in Mollejones allowed us to deeply understand the community's needs in a respectful and immersive way. One recurring issue was the lack of protection from heavy rain, which would flood "La Cancha". Driven by our guiding principles, we created a conceptual solution to address this problem while preserving the integrity of the existing structure and minimizing disturbances.

The average wind direction in Turrialba comes from the east throughout the year and the rain varies with the seasons with about 7.1 months of wet season from May to December. The Sun rises in the East and sets in the West, so the community center only gets direct sunlight on the inside of the court during the times of the sun setting because of the mountain on the East of the building blocking the sun from hitting it at all times of the day.

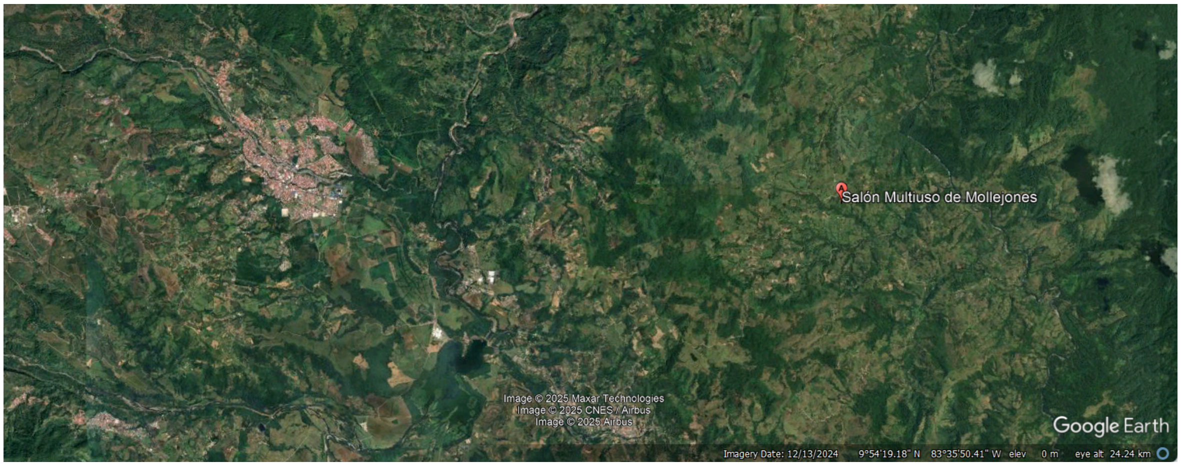


Figure 2. Location of Los Mollejones and their community center, in relation to Turrialba city (left)

Figure 3. General site map of Mollejones relative to the community center, highlighted in red.

Need/Vision Statement

Project Goals:

Short Term: Upgrading the Soccer court's outer shell will prevent rain from entering deep into the area, making its use possible during rainy weather and turning it into a possible shelter for users during a storm, namely, children using the soccer court during the day.

Medium Term: Achievable additions to the community space could include the first steps towards expanding the specialized areas required for new site activities, maintaining the design language of Phase 1 to continue its focus on bio-construction.

Long Term: El Bambú's flat, protected soccer court would serve as an open floor plan that can be used in several ways, for a few hours at a time on specific days. Specialized functions, like rice shucking and fuel storage, will be done on new, bio-constructed spaces which use Phase 1's framework as an anchor and reinforce the status of Los Mollejones as an example in sustainability.

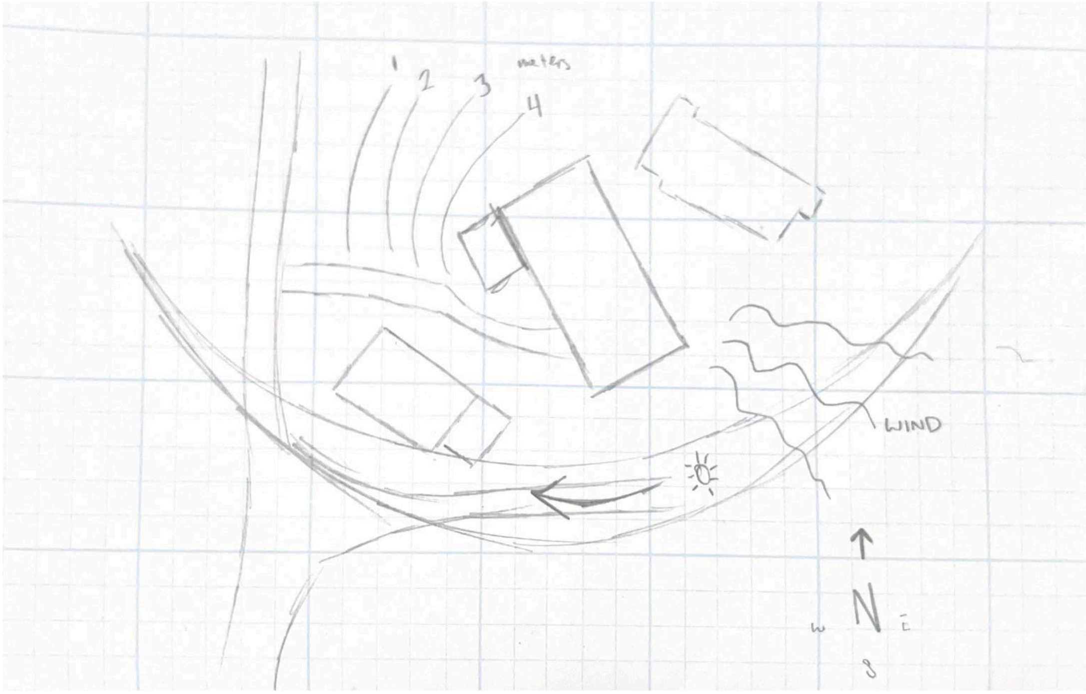


Figure 4. Diagram with the sun path and main wind directions for the area around the Soccer Court..

Guiding Principles

Innovation & Transferability:

Inspired by permaculture, bio construction integrates local resources to minimize environmental waste and reduce the level of disturbance around the intended site. This project leverages natural materials to construct *louver* windows for the space between the double roof and exterior arches on the community center. Bamboo is the proposed medium for the individual *louver* panes as it can be easily sourced and is highly abundant. Slanted downwards at a 30° angle, the *louvers* replicate a common technique seen in the windows of Costa Rican architecture where individual panes allow users to redirect air flow and mitigate interior temperatures. As a permanent fixture, this addition to the community center can be implemented in a similar fashion to other buildings around Mollejones where similar needs have been identified.

Figure 4. Playing Volleyball with the children at the soccer court

Figure 5. The inside of the Mollejones Soccer court

Ethical Standards & Social Equity:

After spending time in this space, our group recognized how important the community center is to residents of Mollejones. With a large gathering space that houses a public kitchen for meals and seating for the mentioned tourist workshops plus multiple bathrooms and storage spaces, we want to ensure that the community can utilize the community center year-round. Weather is the current limitation for use as heavy rain leaks through the double roof and exterior openings; thus, adding the bamboo *louvers* hinders the overflow rain from entering while also providing shade so the community can decide when to use this space rather than the environment. Equity is at the core of our project because sustainability is a movement developed by people for people. Generating more opportunities for the community center to be used despite landscape conditions emphasizes the key role this space plays in neighborhood morality. Locally sourced bamboo encourages more partnerships within Costa Rica while reinvesting in Mollejones' economy via educational ecotourism.

Environmental Quality & Resource Efficiency:

As previously mentioned, our *louver* windows are based on bio construction techniques to lessen the ecological impact of our addition to the community center. Bamboo has rapid growth periods that are enhanced by Costa Rica's tropical climate and is known for its durability in construction. Its lightweight yet non-permeable composition makes transportation easy and with CATIE's help, a natural sealant for preservation can be painted on the bamboo before it's fixed to the community center. For example, Dr Miguel Obregón Gómez has pioneered this type of technology in Costa Rica through his research in pest control and microorganisms. His lab engineered a chemical solution that fertilizes bamboo with a fungus that is absorbed into the plants' walls and kills intruding insects during maturation (Laboratorios Dr Obregón). This would increase resilience and resistance of the bamboo used in our proposed *louver* windows by protecting against diseases or other confounding factors so we recommend CATIE reference this lab or something similar if partnership is documented. Additionally, our project adds to a pre-existing building in Mollejones resulting in minimal interruptions for neighboring wildlife and fewer material demands due to the design.

Economic Performance & Compatibility:

The challenging aspect of going from a conceptual idea to a reality is funding. To address this, we considered leveraging partnerships with big businesses. For example, Rawlings and Firestone, both well-known names in the area, could be approached with a mutually beneficial proposal to secure project funding. Additionally, potential collaborations with Catie and Brian Erickson present great opportunities. CATIE, which has a lot of bamboo available on campus, could provide the primary material, while Brian's expertise in sustainable treatment methods could ensure the material's viability. CATIE could further extend this collaboration by adopting the bamboo structure as a long-term research initiative. By monitoring its performance and compatibility over time, CATIE could evaluate its economic and environmental sustainability, contributing valuable insights to the field.

Contextual & Aesthetic Impact:

To offer the existing structure protection from the rain, the design uses the current metal columns which hold the roof up as the anchoring point for a new, light structure out of bamboo. An outer frame is drilled into the metal columns, from which the *louvers* are fastened into at an angle – enough to let the wind pass as well as indirect light, but also to stop the wind from blowing the rain too deep into the soccer court during a storm. A good angle that should reliably block out the rain is about 30°, but with a closer look at how Mollejones experiences rain may provide a more definitive answer. The new structure starts at the roof and ends 9 feet above the ground to allow people and small-to-medium vehicles to pass under. Because the soccer court is protected from the West by a natural slope and from the North by the kitchen building, only the South and East faces will require this intervention. Similarly, the same *louver* system will be implemented on the secondary roof system intended to dissipate heat which has an opening of approximately 1 ft. Bamboo is a great natural element to soften the room, especially one with steel and concrete like the community center, and provide acoustics to mitigate noise.

Case Studies

Our design is inspired by a few sustainable designs such as the No-footprint house and a new form of treatment for Bamboo.

No Footprint House:

The No Footprint House designed by A01 utilizes the sustainable qualities of bamboo in construction as well as having a passive climate control system that uses the sun and wind to its advantage (ArchDaily 2019). The building's main attribute are its sunshade walls where the walls are basically made out of blinds/shaders to protect from the sun and water while still being open for the sunlight to get it for natural lighting. We are using the water blocking attributes of this design by integrating the sunshade walls into our project in order to be able to mitigate the water that gets into the community center while still letting in the sunlight. Along with that the sunshade walls will help with heat mitigation to make the community center more comfortable.

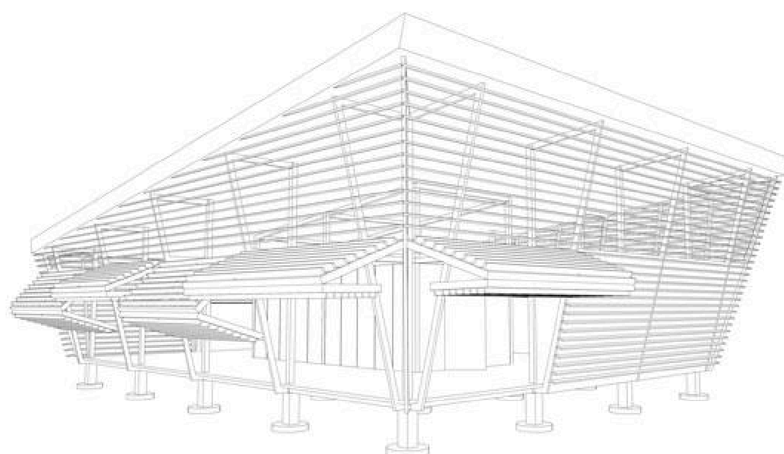


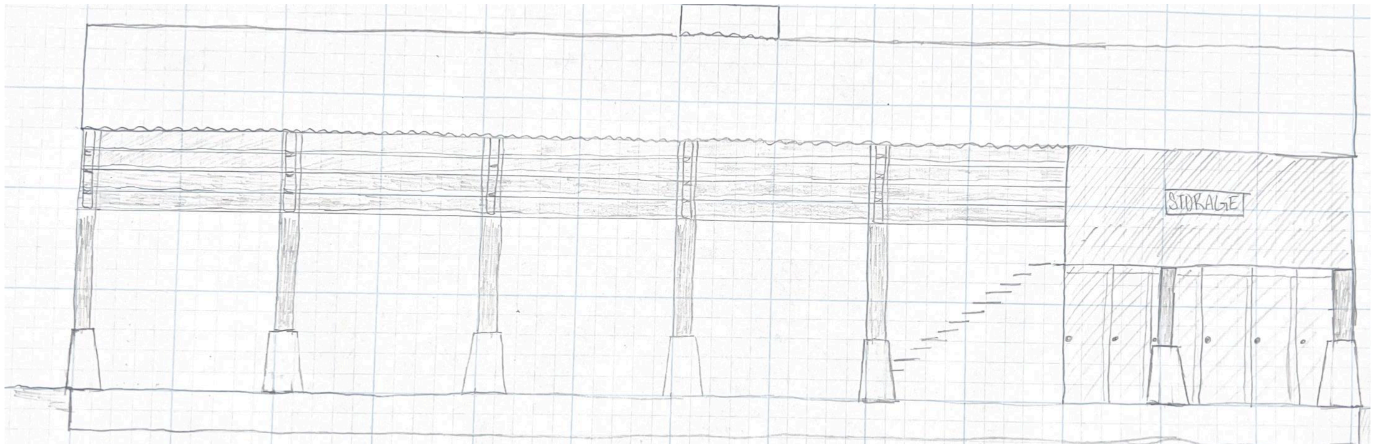
Figure 7. No Footprints House diagram

Sustainable Treatment of Bamboo:

The case study on sustainable treatments of bamboo by Amit Sain, Arun Gaur, Jeetendra Singh Khichad, and Prakash Somani goes over a few different ways to sustainably treat bamboo for construction in ways that don't hurt the environment and can greatly improve building life. This paper recommends certain types/brands of synthetic rubber spray and synthetic resin spray which both increase durability, provide a protective barrier, increase life span, and prevent fungus and disease in the bamboo. These provided options are sustainable and environmentally friendly because they use environmentally friendly, reusable substances and avoid synthetic chemicals (Amit Sain et al 2024 IOP Conf.). These synthetic compounds are safe for the environment and encourage continued sustainability which is what we are going for in this project and thus will be incorporated into our design.

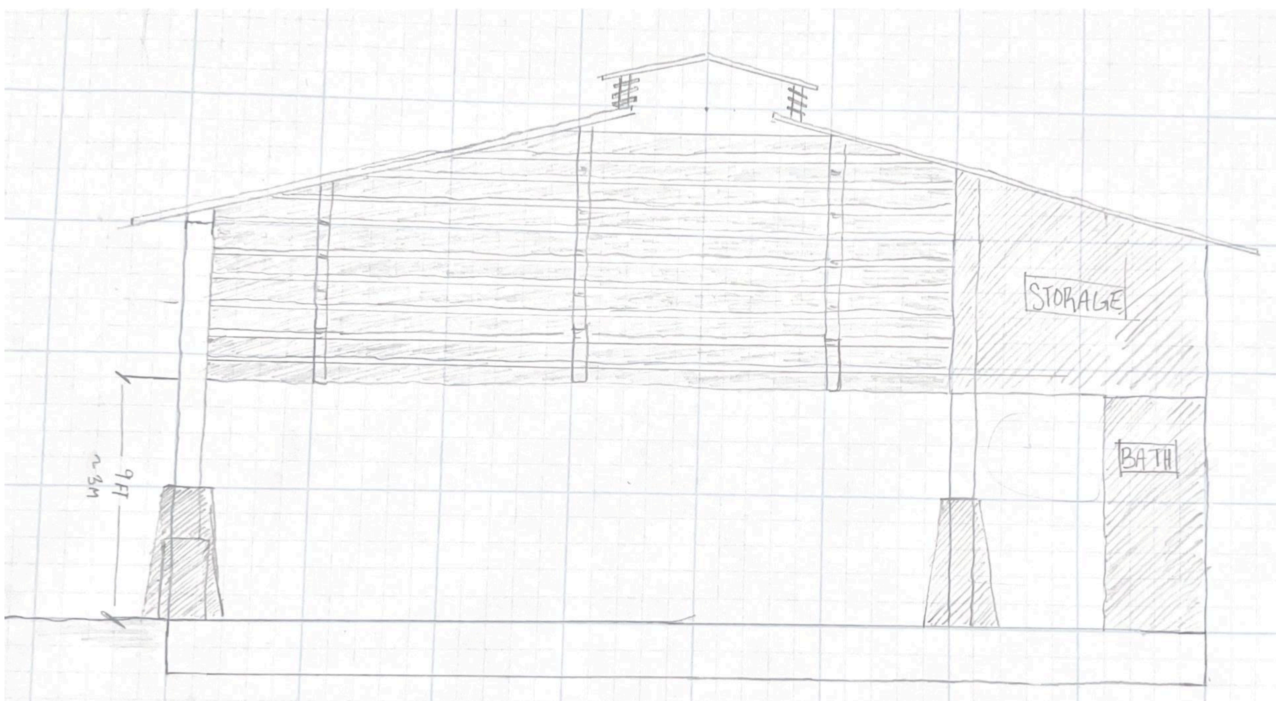
Project Development

West View Elevation:



Drawing Scale: 1:100. The space between the columns is approximately 10ft

North View Elevation:



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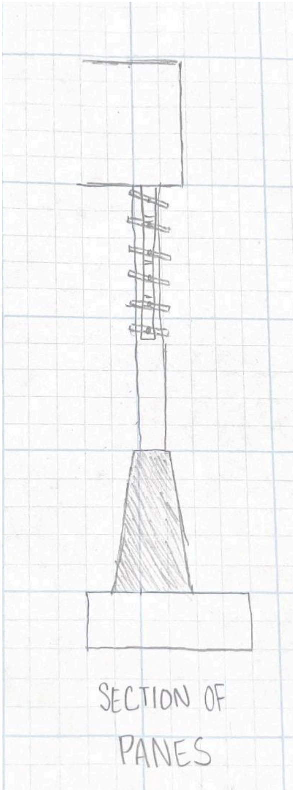
Cross-Section of Panels:



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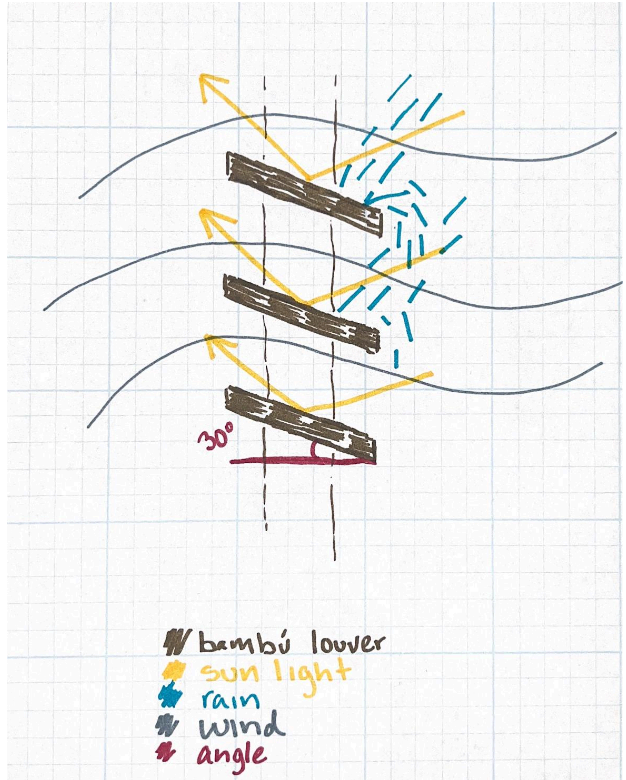
Wind/Light/Rain Diagram:



Graphic scale.

Materials List, Cost Estimate and Funding

This is an approximate list of materials and cost for the project based on sources such as Home Depot, Brian Erickson's Business in Bamboo, and the local Costa Rican hardware store LaFerretería CR. This does not include PPE due to not having an ability to estimate how many people will be working on the project.

Table 1. Itemized cost estimate of the materials needed for this project.

Material	Approximate Quantity	Unit	Material Cost (USD)	Total Cost (USD)
Bamboo	80	Chutes (approx 6 ft)	$5.95/per chute	476
Drills	1	N/A	$90.24 per drill	90.24
Cobalt Drill Bit	1	N/A	$26.87/per bit	26.87
ladder	1	N/A	$179.00/per ladder	179.00
Saws	5	N/A	$13.79/per saw	68.95

Sandpaper	10	Sheets	$0.88/per sheet	8.80
Wood Glue	50	Fl Oz	$0.87/per ounce	43.50
Sealant	600	Fl Oz	$0.90/per ounce	540.00
Paint	300	Fl Oz	$0.55/per ounce	165.00
Mallets	2	N/A	$10.00/per mallet	20.00
-	-	-	-	$1618.36

Feasibility Statement:

Based on our research this project is feasible and possible to do, due to its reasonable costs, simple design, and minimal materials needed. Though due to many gaps in our knowledge and lack of experience in the pricing of materials in Costa Rica there may be some inaccuracies in the costs of items. Another gap in our concept is that our idea may not work due to us not having the ability to do rain studies and wind direction studies but in theory the project should be effective in mitigating the amount of rain that seeps into the common space based on our personal experience when we were in Mollejones.

Overall the project has a simple design that could easily be done by volunteer workers with minimal training which greatly reduces the costs of the project. We can also further reduce the costs of the project by renting tools and partnering with Catie to obtain sustainable bamboo.

Meet Group 3!

From left to right:

- **Ella Snyder:** Colorado State University, Sophomore, Interior Architecture and Design Major, Global Environmental Sustainability Minor
- **Emma Pingel:** Colorado State University, Senior, Ecosystem Science & Sustainability, Minors in Global Environmental Sustainability and Applied Environmental Policy Analysis.
- **Kaylyn Barlow:** Colorado State University Student, Freshman. Construction Management Major.
- **Rafael Serrano Talavera:** Colorado State University, Sophomore, Construction Management Major, Real Estate Minor
- **Roberto Durán Brenes:** Universidad de Costa Rica, Licenciatura en Arquitectura

Sources

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